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Zero-Shot Cross-Domain Transfer via VSA Structural Alignment and Hypervector Similarity Clustering

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Abstract

We present a zero-shot cross-domain knowledge transfer mechanism that operates entirely within a Vector Symbolic Architecture (VSA) framework. By comparing 10,240-bit binary hypervectors of domain concepts, the system automatically discovers structural correspondences between domains and transfers rules without any target-domain training data. The mechanism implements Gentner’s Structure Mapping Theory in binary VSA: domain concepts are encoded as hypervectors, structural similarity is measured via normalised Hamming distance, and rules are lifted from source to target domain through concept substitution.

The system consists of three components: (1) `auto_discover_abstractions()` — an automatic cross-domain concept alignment algorithm using greedy maximum-weight matching over hypervector similarity; (2) `transfer_rule()` — a rule lifting mechanism that substitutes grounded concepts for abstract ones; and (3) provenance tracking — transferred knowledge is tagged with source domain and confidence scores that decay with structural distance.

Evaluation on a balancer→catcher game transfer shows +14% success rate over random baseline with zero target-domain training data. We document honest limitations: the mechanism requires structural isomorphism between domains, HV similarity has a ~0.50 baseline for unrelated concepts making threshold tuning critical, and transfer quality degrades with sparse source-domain rules.

Keywords: Transfer Learning, Zero-Shot Learning, Vector Symbolic Architecture, Structural Analogy, Cross-Domain Knowledge Transfer, Cognitive Architecture, Hyperdimensional Computing

1. Introduction

1.1 The Transfer Problem

A fundamental challenge in AI is enabling agents to apply knowledge learned in one domain to a structurally similar but superficially different domain. Humans do this naturally — a child who learns to balance objects can transfer that understanding to catching objects, recognising the shared structure of “adjust position to intercept a moving target.” Neural transfer learning typically requires hundreds of target-domain samples for fine-tuning [1]. Domain adaptation techniques require access to target distributions [2]. Neither achieves true zero-shot structural transfer.

1.2 Gentner’s Structure Mapping Theory

Gentner [3] proposed that analogical reasoning preserves **relational structure** rather than surface features. When a person sees the analogy between the solar system and an atom, they map “sun orbits planets” to “nucleus orbits electrons” — preserving the relational structure (orbits) while ignoring surface properties (mass, charge). This structure-preserving transfer is the mechanism that allows humans to generalise across radically different domains.

1.3 Our Approach

We implement Gentner’s theory in a Vector Symbolic Architecture (VSA) framework [4]. The key insight: in a 10,240-bit binary hypervector space, concepts with similar relational structure will have measurably higher Hamming similarity than unrelated concepts — even across domains. This

is because VSA role-filler encoding (Section 2) captures relational structure in the hypervector itself.

Our system: 1. Encodes all domain concepts as 10,240-bit binary hypervectors using role-filler binding 2. Computes cross-domain similarity matrices using normalised Hamming distance 3. Discovers structural correspondences via greedy maximum-weight matching 4. Lifts rules from source to target domain through concept substitution 5. Tags transferred knowledge with provenance and confidence scores

All of this operates without any target-domain training data — true zero-shot transfer based on structural alignment.

1.4 Contributions

1. **auto_discover_abstractions()** — an automatic cross-domain concept alignment algorithm using HV similarity with name-based boosting.
2. **Rule lifting** — transferred rules apply in the target domain through abstract concept substitution, with confidence discounting.
3. **Provenance tracking** — transferred knowledge does not overwrite target-domain knowledge and includes source-domain attribution.
4. **Honest evaluation** — including a cold-start analysis showing the mechanism’s dependence on source-domain rule quality.

2. Background

2.1 Vector Symbolic Architecture

VSA [4, 5] encodes knowledge as high-dimensional binary vectors with three algebraic operations:

- **Binding (XOR):** $A \oplus B = A \oplus B$ — creates a representation that is quasi-orthogonal to both operands
- **Bundling (majority vote):** Superposition of multiple vectors — result is similar to all components
- **Permutation (circular shift):** $\hat{k}(v)$ — encodes sequential position

Role-filler encoding:

$$\text{HV}(\text{concept}) = \text{HV}(\text{name}) \oplus \bigoplus_{(\text{role}, \text{filler})} \text{HV}(\text{role}) \oplus \text{HV}(\text{filler})$$

This encoding captures relational structure: two concepts with the same roles and similar fillers will have similar hypervectors, even if they are from different domains. For example, `AGENT_MOVES_TO_TARGET` in a navigation domain and `AGENT_ADJUSTS_TO_INTERCEPT` in a game domain will share structural similarity in their HV representations if their roles (agent, target, action) are encoded similarly.

2.2 Structural Similarity in High Dimensions

In a d-dimensional binary space, the normalised Hamming similarity between two random vectors is:

$$\mathbb{E}[\text{sim}(\mathbf{A}, \mathbf{B})] = 0.5, \quad \sigma \approx \frac{1}{2\sqrt{d}} \approx 0.00494 \text{ (for } d = 10,240\text{)}$$

This means structurally unrelated concepts will have similarity 0.50 ± 0.005 . Any similarity significantly above 0.50 indicates shared structure. The **concentration of measure** phenomenon in high dimensions means this threshold is remarkably sharp — there is very little probability mass between “unrelated” (0.50) and “meaningfully similar” (> 0.55).

2.3 Structure Mapping Theory

Gentner [3] proposed three principles for analogical mapping: 1. **Relational focus:** Map relations (predicates), not attributes 2. **Systematicity:** Prefer mappings that preserve higher-order relational structure 3. **One-to-one mapping:** Each element in the source maps to at most one element in the target

Our implementation follows all three principles: role-filler encoding captures relations (1); HV similarity naturally favours systematic mappings because structured similarity compounds (2); and greedy matching enforces one-to-one correspondence (3).

3. The Transfer Mechanism

3.1 Domain Representation

Each domain D is represented as: - A set of **concept HVs**: $\{\text{HV}(c), \text{HV}(c), \dots, \text{HV}(c)\}$ — each concept encoded as a 10,240-bit binary hypervector - A **rule base**: $\{R, R, \dots, R\}$ — symbolic rules over domain predicates (learned by the RuleLearner from experience) - A **causal graph**: directed graph of cause-effect relationships (learned by CausalDiscovery)

3.2 Auto-Discovery of Abstract Concepts

Algorithm 1: `auto_discover_abstractions(domain_a, domain_b)`

From the AnalogyEngine (analogy.py, 609 lines):

```
INPUT: concepts_a = {(name, HV)} from domain_a
       concepts_b = {(name, HV)} from domain_b
       similarity_threshold = 0.52
```

```
1. Build similarity matrix S:
   for each (a_name, a_hv) in concepts_a:
     for each (b_name, b_hv) in concepts_b:
       S[a, b] = sim(a_hv, b_hv)

   # Name-based boosting:
   if shared_prefix(a_name, b_name) > 3 chars:
     S[a, b] += 0.05
   if a_name in b_name or b_name in a_name:
     S[a, b] += 0.08
```

2. Greedy 1-to-1 matching (descending similarity):


```

matched_a = {}
matched_b = {}
pairs = sort all (a, b, S[a,b]) by S[a,b] descending

for each (a, b, score) in pairs:
    if a not in matched_a and b not in matched_b:
        if score > similarity_threshold:
            matched_a.add(a)
            matched_b.add(b)
            register AbstractConcept:
                name = "AUTO_{a_name}_{b_name}"
                maps_to = {domain_a: a, domain_b: b}
                similarity = score
      
```
3. Invalidate cached analogies between domains
4. Return discovered abstract concepts

Why greedy matching? Optimal bipartite matching (Hungarian algorithm) is $O(n^3)$, but in practice the number of concepts per domain is small (< 100), and greedy matching produces near-optimal results because HV similarity is sharply concentrated around 0.50 — pairs that should match have significantly higher similarity.

Why the similarity threshold of 0.52? The expected similarity between unrelated concepts is 0.50 with $\sigma = 0.005$. A threshold of 0.52 is approximately 4 σ above random, corresponding to $p < 0.0001$ for a false positive. However, this threshold is deliberately conservative — it may miss weak structural correspondences. The name-based boosting compensates by lifting pairs with lexical similarity above the threshold.

Maximum-entropy adaptive threshold. The hard-coded 0.52 can be replaced with a principled, adaptive threshold derived from Jaynes’ maximum-entropy principle [15]. Given the null distribution of random HV similarity ($\mu = 0.5$, $\sigma = 1/(2\sqrt{d})$) and an expected match fraction π (the prior probability that a random concept pair is a true correspondence), the optimal threshold is:

$$\theta^* = \mu + \Phi^{-1}(1 - \pi) \cdot \sigma$$

where Φ^{-1} is the standard normal quantile function. For $d = 10,240$ and $\pi = 0.1$: $\theta^* = 0.506$. This is the threshold that maximizes entropy subject to the constraint that only a fraction π of random pairs should exceed it — the least-informative threshold consistent with the desired selectivity. The `max_entropy_threshold()` method in `AnalogyEngine` computes this adaptively from the VSA dimensionality and domain statistics.

3.3 Rule Lifting and Application

Algorithm 2: `transfer_rule(rule, source_domain, target_domain)`

```

INPUT: rule = {conditions: [pred, pred, ...],
               action: act,
               confidence: conf}
         abstract_mapping = discovered correspondences
      
```

1. For each predicate in rule.conditions:
 - a. Find abstract concept that maps predicate in source_domain
 - b. If found: substitute with target_domain grounding
 - c. If not found: attempt partial name matching
 - d. If still not found: mark predicate as UNGROUNDED
2. For rule.action:
 - a. Same mapping procedure
 - b. If action cannot be grounded: transfer fails
3. Compute transferred confidence:


```
transferred_conf = rule.confidence * min(mapping_similarities)
```
4. Return TransferredRule:


```
conditions: [mapped_pred , mapped_pred , ...]
action: mapped_act
confidence: transferred_conf
source: source_domain
provenance: "Transferred from {source} via structural alignment"
```

Confidence discounting: The transferred rule’s confidence is the product of the original rule confidence and the minimum concept-mapping similarity. This means weak mappings produce low-confidence transfers, which will lose in GWT coalition competition against rules with direct evidence.

3.4 Zero-Shot Action Selection

When the agent encounters a target-domain state with no learned rules:

- ```
zero_shot_action(state, target_domain):
```
1. For each source\_domain with learned rules:
    - a. Find mappings to target\_domain (from auto\_discover or cached)
    - b. For each source rule matching the mapped state:
 

```
transfer_rule(rule, source, target)
```
    - c. Add transferred rules as GWT coalitions
  2. GWT competition selects the best transferred rule
  3. Execute with provenance tagging

### 3.5 Provenance and Non-Interference

Transferred knowledge follows strict isolation principles: - Tagged with source\_domain identifier - Does not overwrite existing target-domain rules - Confidence tracks transfer reliability and decays over time - If a transferred rule is contradicted by target-domain experience, it is demoted (not the source rule)

### 3.6 Transfer Quality: Functoriality Score

A structural mapping between domains is, in the language of category theory, a **functor**  $F: C \rightarrow D$  — a structure-preserving map between categories [13, 14]. A faithful functor preserves the “distinctness” of morphisms: objects that are far apart in the source should remain far apart in the target, and objects that are close should remain close. We measure how well our discovered mapping satisfies this requirement with a **functoriality score**:

$$F = 1 - \frac{1}{\binom{n}{2}} \sum_{i < j} |\text{sim}(a_i, a_j) - \text{sim}(F(a_i), F(a_j))|$$

where  $\{a_1, \dots, a_n\}$  are the source-domain concepts participating in the mapping and  $F(a_i)$  is the corresponding target-domain concept. The score ranges from 0 to 1: a perfect isomorphic mapping (where all pairwise distances are preserved exactly) yields  $F = 1.0$ , while a random mapping yields  $F = 0.5$  in high dimensions.

This metric directly operationalises Gentner’s systematicity principle [3]: good analogies preserve relational structure, and functoriality measures exactly how much structure is preserved. In our implementation, `functoriality_score()` in `AnalogyEngine` computes this over all mapped concept pairs after auto-discovery. In the balancer→catcher experiment (Section 5.1), the functoriality score is 1.0, confirming that the discovered mapping is a perfect structural isomorphism.

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## 4. Theoretical Analysis

### 4.1 When Does Transfer Work?

**Theorem (informal):** Transfer succeeds when the source and target domains share structural isomorphism — i.e., there exists a mapping  $M: \text{concepts\_source} \rightarrow \text{concepts\_target}$  such that:

$$\forall \text{ rule } R = (C \rightarrow A) \text{ in source: } M(C) \rightarrow M(A) \text{ is valid in target}$$

**Sufficient condition for discovery:** If the role-filler structure of corresponding concepts produces HV similarity above the threshold:

$$\text{sim}(\text{HV}(c_s), \text{HV}(c_t)) > \theta = 0.52$$

then the greedy matching algorithm will discover the correspondence with high probability.

### 4.2 When Does Transfer Fail?

Transfer fails in three cases:

1. **No structural isomorphism:** The domains are truly unrelated. All cross-domain similarities are  $\leq 0.50$ , and no pairs exceed the threshold.
2. **Threshold miscalibration:** If the threshold is too high, valid correspondences are missed. If too low, spurious correspondences are discovered.

3. **Sparse source rules:** Even if structural correspondences are found, if the source domain has few learned rules, there is little to transfer. This is the **cold-start problem** observed in our experiments.

### 4.3 Concentration of Measure

In 10,240-dimensional binary space, the Hamming distance between random vectors is tightly concentrated around  $d/2$ . The probability that two random vectors have similarity  $> 0.52$  is:

$$P(\text{sim} > 0.52) \approx 1 - \Phi\left(\frac{0.52 - 0.50}{0.00494}\right) = 1 - \Phi(4.05) \approx 0.000026$$

where  $\Phi$  is the standard normal CDF. This means false positive correspondences are extremely rare in high dimensions — a strong property for automatic discovery.

The maximum-entropy adaptive threshold (Section 3.2) provides the formal connection:  $\tau$  is exactly the  $(1 - \alpha)$ -quantile of this null distribution, derived from the concentration of measure phenomenon rather than chosen ad hoc. This grounds the threshold selection in the same distributional facts that guarantee the rarity of false positives.

## 5. Experiments

### 5.1 Game Transfer: Balancer $\rightarrow$ Catcher

**Source domain (Balancer):** An agent learns to balance a pole by adjusting position. Predicates: {pole\_tilting\_left, pole\_tilting\_right, pole\_centered, agent\_at\_left, agent\_at\_right, agent\_at\_center}. Actions: {move\_left, move\_right, stay}.

**Target domain (Catcher):** An agent must catch falling objects by positioning itself. Predicates: {object\_falling\_left, object\_falling\_right, object\_centered, agent\_at\_left, agent\_at\_right, agent\_at\_center}. Actions: {move\_left, move\_right, stay}.

**Structural correspondence:** pole\_tilting\_left  $\leftrightarrow$  object\_falling\_left, pole\_tilting\_right  $\leftrightarrow$  object\_falling\_right, etc. The agent position predicates are shared.

#### Results:

| Metric                       | Random Baseline | With Transfer      | Improvement |
|------------------------------|-----------------|--------------------|-------------|
| Success rate                 | 33%             | <b>47%</b>         | <b>+14%</b> |
| Average score                | 0.0             | <b>+0.49</b>       | +0.49       |
| Rules transferred            | —               | 4                  | —           |
| Abstract concepts discovered | —               | 5                  | —           |
| Transfer time                | —               | 0.81 ms (Rust VSA) | —           |
| Functoriality score          | —               | 0.995              | —           |

The +14% improvement over random with zero target-domain training data demonstrates that structural alignment can provide a meaningful cold-start advantage.



## 5.2 Dissimilar Domain Transfer: Navigation    Trading

**Source domain:** Robot navigation with predicates {obstacle\_ahead, goal\_visible, path\_clear, ...}.

**Target domain:** Financial trading with predicates {price\_rising, volume\_high, trend\_bullish, ...}.

These domains share no surface features. The structural correspondences are weaker:

| Metric                         | Result                             |
|--------------------------------|------------------------------------|
| Abstract concepts discovered   | 2 (out of ~10 per domain)          |
| Rules successfully transferred | 1                                  |
| Transfer confidence            | 0.31 (below typical GWT threshold) |

**Interpretation:** The system correctly identifies that these domains have minimal structural overlap. The 2 discovered correspondences are weak (similarity barely above 0.52), and the single transferred rule has such low confidence that it would lose in GWT competition against even a moderately confident memory-based coalition. This is the **correct behaviour** — the system should not force transfer between unrelated domains.

## 5.3 Cold-Start Degradation

We measured transfer performance as a function of source-domain training episodes:

| Source Episodes | Rules Learned | Rules Transferred | Target Success |
|-----------------|---------------|-------------------|----------------|
| 0               | 0             | 0                 | 33% (random)   |
| 10              | 1             | 1                 | 36%            |
| 50              | 3             | 2                 | 41%            |
| 100             | 5             | 3                 | 44%            |
| 500             | 9             | 4                 | 47%            |

**Observation:** Transfer quality scales with source-domain rule quality. With zero source rules, transfer equals random (0% improvement). This confirms the fundamental dependence on source-domain learning quality — transfer cannot create knowledge, only move it.

## 6. Comparison with Existing Approaches

| Approach                 | Target Samples | Structural? | Automatic? | Interpretable? |
|--------------------------|----------------|-------------|------------|----------------|
| Fine-tuning (neural) [1] | 100+           | No          | Yes        | No             |
| Domain adaptation [2]    | Unlabelled     | No          | Yes        | No             |
| Hand-coded domain maps   | 0              | Yes         | <b>No</b>  | Yes            |
| Meta-learning (MAML) [6] | 5–10           | No          | Yes        | No             |

| Approach                | Target Samples | Structural? | Automatic? | Interpretable? |
|-------------------------|----------------|-------------|------------|----------------|
| <b>NSCK (this work)</b> | <b>0</b>       | <b>Yes</b>  | <b>Yes</b> | <b>Yes</b>     |

NSCK is the only approach that achieves zero target-domain samples, automatic structural mapping, and full interpretability. The trade-off is that it requires structural isomorphism between domains — a stronger assumption than statistical distribution similarity.

## 7. Limitations

We document the following limitations honestly:

1. **Binary HV similarity has ~0.50 baseline for unrelated concepts.** The gap between “unrelated” (0.50) and “meaningfully similar” (0.52+) is only 4%. In noisy environments or with many concepts, false positives become more likely. The maximum-entropy adaptive threshold (Section 3.2) provides principled, dimension-aware tuning, but the fundamental narrowness of the gap remains a challenge.
2. **Works for symbolic domains only.** The transfer mechanism operates on symbolic predicates and rules. Raw sensor domains (pixels, audio) require a grounding layer to convert continuous observations to discrete predicates before transfer can operate.
3. **Transfer quality degrades with sparse source rules.** If the source domain has few learned rules (cold start), there is little to transfer. The mechanism cannot create new knowledge — it can only move existing knowledge across domains.
4. **Greedy matching is suboptimal.** The greedy 1-to-1 matching algorithm does not guarantee the global optimum. For domains with many near-threshold similarities, Hungarian matching would produce better alignments at  $O(n^3)$  cost. The functoriality score (Section 3.6) provides a post-hoc quality measure — a low score after greedy matching would indicate that a more expensive algorithm is warranted.
5. **Name-based boosting introduces bias.** The +0.05/+0.08 name-similarity boost favours domains with similar naming conventions. Completely independently named but structurally identical domains may be missed.
6. **Only tested on simple game domains.** The evaluation uses toy domains with  $< 10$  predicates each. Scaling to complex real-world domains with hundreds of predicates is untested.

## 8. Future Work

1. **Large-scale structural alignment.** Test on domains with 100+ predicates using approximate HV similarity search (NSW index or HNSW).
2. **Hierarchical transfer.** Discover not just concept correspondences but hierarchical relational structures (relations between relations).

3. **Transfer with learned grounding.** Combine SNN perception layer output with transfer — so agents can transfer from a domain where they have sensor grounding to a domain where they do not.
4. **Multi-source transfer.** Aggregate correspondences from multiple source domains for more robust target-domain bootstrapping.
5. **Formal verification.** Prove conditions under which greedy matching produces optimal or near-optimal alignments in VSA.
6. **Categorical analysis of transfer conditions.** Formalise the relationship between VSA structural alignment and category-theoretic functors. The functoriality score (Section 3.6) is a first step; a full categorical treatment would characterise when domain transfer preserves compositional structure (natural transformations) and when it does not, potentially yielding necessary and sufficient conditions for successful transfer.
7. ~~**Hierarchical resonator decoding for transfer.**~~ **Done (V8)** — `HierarchicalResonatorNetwork` now provides 2-level (L1 sentence-role + L2 clause-role) factorization, enabling structural alignment at multiple levels of abstraction. The `factorize_hierarchical(hv, depth=2)` method returns a `{"L1": ..., "L2": ...}` dict that can be used as the alignment substrate for cross-domain transfer.

V13 adds `PatternGeneralizer` (`learning/pattern_generalizer.py`) for automatic abstraction-level estimation: `level = min(1, count / (min_members · 4))`. `CrossModalAssociativeMemory` (`memory/cross_modal_associative_memory.py`) enables cross-modal entity binding via XOR, allowing transfer across perception modalities. The full test suite now includes 1,437 collected tests.

**Current test suite (V13, February 2026): 1,424 passed, 7 skipped, 4 xfailed.**

---

## 9. Conclusion

We have presented a zero-shot cross-domain knowledge transfer mechanism operating entirely within a Vector Symbolic Architecture framework. The system: - Automatically discovers structural correspondences between domains by comparing 10,240-bit binary hypervectors - Transfers rules from source to target domain through concept substitution - Achieves +14% success rate over random baseline with zero target-domain training data - Correctly refrains from forced transfer between structurally dissimilar domains - Tags all transferred knowledge with provenance and confidence scores

The mechanism implements Gentner’s Structure Mapping Theory using the mathematical properties of high-dimensional binary vectors — concentration of measure ensures that structural correspondences produce measurably higher similarity than noise, while the self-inverse property of XOR binding enables clean concept substitution.

The honest limitations are clear: dependence on structural isomorphism, sensitivity to threshold tuning, degradation with sparse source rules, and evaluation limited to toy domains. We present this as a demonstration that principled structural transfer is achievable within a lightweight, interpretable VSA framework — and invite the community to test these ideas at scale.

Code and benchmarks: <https://github.com/shiva2321/Neuro-Symbolic-Cognitive-Kernel>

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*All results are derived from the actual NSCK implementation. The +14% transfer result comes from the `benchmark_report.md` evaluation. We welcome replication, critique, and contribution.*